

Dear Leonardo Brito:

Congratulations - your paper #1571157280 ('GondoCheck: A Vision-Based Backend for Automated Shelf Auditing') for SBrT 2025 has been **accepted** and will be presented in the session titled ____.

The reviews are below or can be found at [1571157280](#).

Detailed Review Form 1

Competence: Please assess your familiarity with the subject matter of the paper.

Familiar with this area of research (2)

Relevance: How relevant is the paper to the scope of SBrT?

Somewhat Relevant (2)

Presentation: Evaluate the quality of the presentation.

Well written (4)

Technical Correctness: Assess the technical content and the soundness of the work.

Good (4)

Technical Correctness: Assess the technical content of the work.

Good (4)

Contribution: Assess the originality of the work and its contribution to the area of research.

Variation of a known concept (2)

Contribution: Considering that this is an *undergraduate* paper, please assess the quality of the work.

Variation of a known concept (2)

Bibliography: Please rate the paper bibliography.

Satisfactory (2)

Recommendation: State your recommendation regarding the acceptance/rejection of the paper.

Weak accept (3)

Strengths: What are the most important reasons to accept this paper?

The paper tackles a practical and interesting problem by proposing GondoCheck, a cloud backend that processes a single gondola photo to detect stocked items, match them to target SKUs using deep metric learning, and deliver real-time shelf-share analytics.

Weaknesses: What are the most important reasons NOT to accept this paper?

The main weakness is the limited validation scope: only six SKUs from the same brand were used to validate the pipeline, whereas real supermarket aisles contain hundreds of variants, raising concerns about scalability and robustness.

Comments: Please provide a brief summary of the contributions and give comments to the authors

Some points require clarification or improvement to strengthen the manuscript:

* The abstract mentions that out-of-stock issues "erode customer loyalty," but the practical motivation for using shelf photos isn't fully clear. Please try to improve why photo-based shelf auditing is needed if stores already track inventory through sale systems. In particular, improve the explanation (maybe in the Introduction section) in which scenarios stock levels in the system don't reflect the actual shelf state (e.g., backroom stock, misplaced items, restocking delays), and why image-based monitoring adds value.

* On page 2, Section III.B, the notation $B=\{b_i\}_{i=1}^M$ is introduced but the meaning of M is not clearly defined. Please specify M corresponds to the number of detected products in the image.

- There is a formatting issue: missing space in "images(products)".
- The manuscript appears to use the symbol " τ " for two different purposes: once to denote the target SKU set (τ), and later as a similarity threshold (e.g., 0.80, 0.90) for embedding comparisons. This reuse is confusing. I recommend using distinct symbols and clarifying the rationale for different threshold values for Euclidean (0.90) and cosine similarity (0.80). In the manuscript is said "If the similarity exceeds a fixed threshold $\tau = 0.80$, the crop is assigned the label of SKU j^* ; otherwise, it remains unlabelled. The value of τ was selected through grid search on a held-out validation subset to optimise macro $F1$." However, I did not find any explanation regarding the 0.90 threshold.
- Can the authors explain why ESRGAN improves results when combined with cosine similarity, but not with Euclidean distance? Could this be related to how these metrics handle feature scaling or magnitude? Additionally, might other factors such as varying distances of photos to products also influence these results?
- Another limitation worth considering (maybe worth to add in the limitation list) is the potential confusion of visually similar products from other brands, especially in cluttered aisle scenarios.

* Why in Equation (5) use $\sum_{i=1}^M 1$? This can be simplified to just M .

Detailed Review Form 2

Competence: Please assess your familiarity with the subject matter of the paper.

Familiar with this area of research (2)

Relevance: How relevant is the paper to the scope of SBrT?

Definitely Relevant (4)

Presentation: Evaluate the quality of the presentation.

Well written (4)

Technical Correctness: Assess the technical content and the soundness of the work.

Good (4)

Technical Correctness: Assess the technical content of the work.

Good (4)

Contribution: Assess the originality of the work and its contribution to the area of research.

Variation of a known concept (2)

Contribution: Considering that this is an *undergraduate* paper, please assess the quality of the work.

Variation of a known concept (2)

Bibliography: Please rate the paper bibliography.

Good (3)

Recommendation: State your recommendation regarding the acceptance/rejection of the paper.

Strong accept (4)

Strengths: What are the most important reasons to accept this paper?

The paper presents a solution to a highly relevant problem in the Brazilian retail sector: the empty shelf spots or rupturas. Applies modern image processing techniques integrated into a web-based system that can be easily incorporated into store staff workflows. The method achieves strong results that validate the potential of the proposed approach.

Weaknesses: What are the most important reasons NOT to accept this paper?

The paper does not provide comparison with existing methods in the literature, nor does it compare its performance with available commercial solutions.

Comments: Please provide a brief summary of the contributions and give comments to the authors

The main contribution of the paper is the development of a complete pipeline for product detection and identification on supermarket shelves using images. The method employs a ResNet-50-based DETR model for product localization, followed by ESRGAN for visual enhancement of detected crops, and finally uses ResNet-152 embeddings with cosine similarity for SKU identification. The solution is implemented on a standard cloud infrastructure (Azure) and provides an end-to-end system capable of delivering real-time shelf-share analytics and visual overlays. An ablation study was also performed to validate the contributions of each component in the pipeline.

The paper is very interesting and well-written. It addresses a highly relevant problem in Brazil, where large supermarket chains are common. The literature review is solid, and the experiments are well-designed and with good results. However, I offer the following comments:

1. In the introduction, it is mentioned that the system returns diagnostic feedback in a matter of seconds. However, no experiments are presented to quantify this latency.
2. In the related work section, the authors state that this paper lies at the intersection of academic and commercial approaches. Therefore, a comparison with published results from the literature would be valuable to position the proposed method more clearly.

That said, the limitations and future work sections are well-articulated and directly address challenges observed during experimentation.

Detailed Review Form 3

Competence: Please assess your familiarity with the subject matter of the paper.

Outside main area of research (0)

Relevance: How relevant is the paper to the scope of SBrT?

Somewhat Relevant (2)

Presentation: Evaluate the quality of the presentation.

Readable (2)

Technical Correctness: Assess the technical content and the soundness of the work.

Flaws but Easy to Correct (2)

Technical Correctness: Assess the technical content of the work.

Several Flaws (0)

Contribution: Assess the originality of the work and its contribution to the area of research.

Contains an original contribution (4)

Contribution: Considering that this is an *undergraduate* paper, please assess the quality of the work.

Variation of a known concept (2)

Bibliography: Please rate the paper bibliography.

Needs improvement (1)

Recommendation: State your recommendation regarding the acceptance/rejection of the paper.

Weak reject (1)

Strengths: What are the most important reasons to accept this paper?

A solução proposta não requer planogramas ou calibração de loja, demonstra robustez ao alcançar 85.70% de acurácia global e 74.90% de macro F1 em 91 imagens reais de supermercados. O trabalho também oferece uma contribuição para a pesquisa, fornecendo um benchmark público conhecido para variantes de detergente líquido brasileiro.

Weaknesses: What are the most important reasons NOT to accept this paper?

O trabalho peca bastante no que tange aspectos técnicos de escrita científica. A escolha de idioma de escrita não fica clara como Inglês Americano ou Inglês Britânico, aparecendo o uso de palavras como "recognize" (American English) e, em outros momentos, "recognise" (British English).

Quanto às referências, há, também, a quebra de padrões. A referência de número 10 não especifica onde o trabalho foi publicado; A de número 12 possui como nome do autor "Isalia99", onde o ideal seria o nome completo do autor; A referência de número 16 possui espaços ao redor dos hífen do link, o que dificulta a consulta por parte do leitor.

Apesar do trabalho se mostrar promissor, a forma da escrita parece mostrar descaso com a magnitude do próprio trabalho, chegando a ter valores de Acurácia e F1-score diferentes da seção resumo para a seção Conclusão.

Comments: Please provide a brief summary of the contributions and give comments to the authors

Prezado(s) autor(es),

Apesar do mérito e do potencial inovador do trabalho apresentado, o mesmo mostra diversas deficiências significativas nos aspectos técnicos da escrita científica e formatação. Há uma inconsistência notável no uso da língua inglesa, alternando entre convenções do inglês americano e britânico (e.g., "recognize" e "recognise"), o que prejudica a padronização. Adicionalmente, a seção de referências necessita de revisão crítica: a referência de número 10 carece de detalhes cruciais sobre sua publicação; a referência de número 12 emprega um username "Isalia99" em vez do nome completo do autor, que seria o ideal em um contexto acadêmico; e a referência de número 16 possui espaços indevidos em sua URL, tornando o link inoperante e dificultando a consulta por parte do leitor. Tais

pontos sugerem um descuido na apresentação formal que não condiz com a relevância do conteúdo técnico do trabalho, chegando ao ponto dos valores de acurácia serem diferentes na seção Resumo para a seção de Conclusão.